Osdag
Open steel design and graphics

OsdagBridge

Open Source Software for Steel Girder Bridge Design

Design Report

Project Name	OsdagBridge report verification - synthetic fixture
Project Location	Test fixture
Author / Designer	Report regression
Reviewer	
Organization	OsdagBridge
Client Name and Organization	Test
Job Number	
Date	2026-11-11
Report Version	Layout and data-mapping test; not a completed engineering design

Table of Contents

Executive Summary 2
Project Overview 2
Key Design Outcomes Summary 3
Design Assumptions and Limitations 3
1 Project Information 4
1.1 Project and Design Team Details 4
1.2 Applicable Codes and Standards 4
2 Input Parameters 5
2.1 Basic Inputs (User-Defined) 5
2.2 Additional Inputs 7
3 Loads and Load Combinations 12
4 Analysis Results 16
5 Design Checks 18
5.1 Plate Girder Design 18
5.2 Deck Slab Design 36
5.3 Cross Bracing Design 41
5.4 End Diaphragm Design 42
5.5 Overall Design Check Summary 43
6 Drawings and Visualizations 45
6.1 Bridge Configuration and Layout 45
6.2 Plate Girder — Detailed Views 45
6.3 Cross Bracing Detail 45
6.4 End Diaphragm Detail 46
7 Material Take-off & Quantity Summary 47
8 Standards & Assumptions 49
8.1 Design Standards 49
8.2 Analysis and Design Assumptions of This Version 51
8.3 Known Limitations of This Version 53
References 54

Executive Summary

This section provides a concise summary of the bridge design, key inputs, governing loads, and final design outcomes.

Project Overview

Bridge Type	Highway Bridge
Design Standard	IRC 5, IRC 6, IRC 22, IRC 24, IS 800
Span	30.0 m
Carriageway Width	7.5 m
No. of Girders	5
Girder Spacing	2.43
Deck Thickness	250.0
Overall Design Status	Fail
Governing Check	Moment
Overall Utilization Ratio (max)	1.250 (Steel plate girders)

[PLACEHOLDER: Figure 1 – Overall Bridge Plan]

Figure 1 – Overall Bridge Plan

[PLACEHOLDER: Figure 2 – Typical Cross-Section (with girder, deck, barriers, footpath)]

Figure 2 – Typical Cross-Section (with girder, deck, barriers, footpath)

[PLACEHOLDER: Figure 3 – 3D View of Bridge Superstructure]

Figure 3 – 3D View of Bridge Superstructure

Table 1 – Final Bridge Geometry (after optimization)

	G1	**G2**	**G3**	**G4**	**G5**
Member ID	G1M1	G2M1	G3M1	G4M1	G5M1
Section Designation					
Governing Check	Moment	Moment	Moment	Moment	Moment
Utilization Ratio	1.210 (FAIL)	1.220 (FAIL)	1.230 (FAIL)	1.240 (FAIL)	1.250 (FAIL)

Note: Utilization ratio (UR) = demand / capacity. A value ≤ 1.0 *indicates a passing check.*

Key Design Outcomes Summary

Steel plate girders: FAIL
Concrete deck slab: PASS
Cross bracing: Not available
End diaphragms: Not available

Design Assumptions and Limitations

- Additional inputs not provided by the user were assumed by software per IRC/IS code defaults or practical consideration.
- Grillage analysis was performed using OSPGrillage assuming simply supported I-girders.
- Substructure and foundation design are not included in this report.
- Splice connections and bearings are not designed in this version.

1 Project Information

This section records all project metadata as entered by the designer.

1.1 Project and Design Team Details

Project Name	OsdagBridge report verification - synthetic fixture
Project Location	Test fixture
Designer	Report regression
Reviewer	
Organization	OsdagBridge
Client	Test
Software Version	OsdagBridge

1.2 Applicable Codes and Standards

- Indian Roads Congress (IRC) 5: General Features of Design
- Indian Roads Congress (IRC) 6: Loads and Load Combinations
- Indian Roads Congress (IRC) 22: Composite Construction (Limit State Design)
- Indian Roads Congress (IRC) 24: Steel Road Bridges (Limit State Method)
- Indian Roads Congress (IRC) 112: Concrete Road Bridges (deck design)
- Indian Roads Congress Special Publication (IRC SP) 114: Seismic Design of Road Bridges
- Indian Standard (IS) 800: General Construction in Steel
- Indian Standard (IS) 2062: Hot Rolled Structural Steel Specification
- Indian Standard (IS) 6006: Steel Bearings

2 Input Parameters

This section documents all inputs provided to OsdagBridge. User-provided inputs are clearly distinguished from software-assumed defaults. Where the user did not supply a value, the software has applied the IRC/IS code default or an empirical guideline; these are annotated with an asterisk (*).

2.1 Basic Inputs (User-Defined)

Note: These inputs are mandatory and were provided by the user.

Table 2.1: Project Location

parameter	value
Project Location	Test fixture
Latitude / Longitude	,
Seismic Zone (IRC 6)	
Basic Wind Speed (IRC 6)	
Shade Temp. Max / Min (IRC 6)	°C / °C

Table 2.2: Bridge Geometry

parameter	value
Type of Structure	Highway Bridge
Span (m)	30.0 m
Carriageway Width (m)	7.5 m
Include Median	No
Footpath	Both
Skew Angle (degrees)	(IRC 24 Cl. 504.8 limit: ±15°)

** Software default value*

Table 2.3: Material Selection

parameter	value
Girder Steel Grade (IS 2062)	E 350A
Cross Bracing Steel Grade	E 350A
End Diaphragm Steel Grade	E 350A
Concrete Deck Grade (IRC 22)	M40

2.2 Additional Inputs

Where the user has modified additional inputs, those values are reported here. Where no modification was made, the software default is shown.

Table 2.4: Typical Section Details

parameter	value
Overall Bridge Width (m)	12.149999999999999
No. of Girders	5
Girder Spacing (m)	2.43 m
Deck Overhang Width (m)	1.215 m
Deck Thickness (mm)	250.0 mm
Footpath Width (m)	1.5 m (IRC 5 Cl. 104.3.6 min: 1.5 m)
No. of Traffic Lanes	(per IRC 5 Cl. 104.3.1)

Table 2.5: Components Details

parameter	value
Crash Barrier Type	IRC 5 - RCC Crash Barrier
Crash Barrier Load (kN/m)	7.557
Railing Type	IRC 5 - RCC Railing
Railing Load (kN/m)	1.4715
Wearing Course Material	Concrete
Wearing Course Thickness (mm)	50.0 mm

Table 2.6: Girder General Information

Girder	Member ID	Design Mode	Girder Type	Girder Symmetry
G1	G1M1	Optimized	Welded	Girder Symmetric
G2	G2M1	Optimized	Welded	Girder Symmetric
G3	G3M1	Optimized	Welded	Girder Symmetric
G4	G4M1	Optimized	Welded	Girder Symmetric
G5	G5M1	Optimized	Welded	Girder Symmetric

Table 2.7: Girder Section Dimensions

Girder	Total Depth, D (mm)	Web, t_w (mm)	Top Flange (b_{tf}, t_{tf}) mm	Bottom Flange (b_{bf}, t_{bf}) mm
G1	1670 mm	All mm	510 mm, All mm	510 mm, All mm
G2	1670 mm	All mm	510 mm, All mm	510 mm, All mm
G3	1670 mm	All mm	510 mm, All mm	510 mm, All mm
G4	1670 mm	All mm	510 mm, All mm	510 mm, All mm
G5	1670 mm	All mm	510 mm, All mm	510 mm, All mm

Table 2.8: Girder Restraint and Stiffener Details

Girder	Torsional / Warping Restraint	Web Philosophy	Intermediate Stiffeners	Longitudinal Stiffeners	Bearing Stiffener
G1	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 2440 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G2	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 2440 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G3	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 2440 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G4	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 2440 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G5	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 2440 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm

Table 2.9: Member Properties: Cross Bracing Details

Location	Member IDs	Type of Bracing	Bracing Section	Spacing (m)
G1 to G2	B1M1 to B1M1	X-Bracing	IS 100 x 100 x 10	15.0 m
G2 to G3	B2M1 to B2M1	X-Bracing	IS 100 x 100 x 10	15.0 m
G3 to G4	B3M1 to B3M1	X-Bracing	IS 100 x 100 x 10	15.0 m
G4 to G5	B4M1 to B4M1	X-Bracing	IS 100 x 100 x 10	15.0 m

Table 2.10: Member Properties: End Diaphragm Details

Location	Member IDs	Type of Bracing	Bracing Section
G1 to G2	E1M1 to E1M2	Cross Bracing	IS 100 x 100 x 10
G2 to G3	E2M1 to E2M2	Cross Bracing	IS 100 x 100 x 10
G3 to G4	E3M1 to E3M2	Cross Bracing	IS 100 x 100 x 10
G4 to G5	E4M1 to E4M2	Cross Bracing	IS 100 x 100 x 10

Table 2.11: Shear Connector Details

parameter	value
Stud Diameter (mm)	
Stud Height (mm)	
Stud f_y (MPa)	
Stud f_u (MPa)	
No. of Studs per Section	

Note: All values are per IRC 22 Table 1 unless user-modified.

Table 2.12: Partial Safety Factors

parameter	value
γ_{M0} (Yielding / Buckling)	1.10
γ_{M1} (Ultimate Stress)	1.25
γ_C (Concrete, Basic)	1.50
γ_s (Reinforcement)	1.15
γ_v (Shear Connectors)	1.25
γ_{fft} (Fatigue Load)	1.00
γ_{Mft} (Fatigue Strength)	1.35

3 Loads and Load Combinations

This chapter summarizes all dead, live, environmental, and thermal loads applied to the bridge structure, together with the design limit state load combinations specified in IRC 6:2017.

Table 3.1: Dead Load – Self Weight Parameters

Parameter	Design Value / Specification
Steel Self-Weight Applied	
Concrete Deck Weight	
Self-Weight Factor	1.00

Table 3.2: Dead Load for Surfacing (DW) & Superimposed Dead Loads

Parameter	Design Value / Specification
Wearing Course Load	Concrete x 50.0
Additional SIDL (Crash Barrier)	7.557 kN/m per barrier
Railing Load	1.4715 kN/m*

Table 3.3: Vehicular Live Load Specifications (IRC 6:2017)

Vehicle Class	Impact multiplier (1 + IF)	Braking Load (kN)	Centrifugal Force (kN)	Selection Status
IRC Class A	1.207	Not available	Not available	Selected per IRC 6 Cl. 204
IRC Class 70R (Wheeled)	1.100	Not available	Not available	Selected per IRC 6 Cl. 204

** Software default value*

Table 3.4: Footway / Footpath Live Load Parameters (IRC 6 Cl. 206)

Parameter	Design Value	Unit	IRC Reference / Description
Applied Pedestrian Live Load Intensity	4.91	kN/m^2	As per IRC 6
Footpath Effective Width	1.50 per side	m	Pedestrian Walkway Width
Uniform Linear Footpath Force	7.36 per side	kN/m	Linear grillage line load

Table 3.5: Wind Load (WL) — per IRC 6

Parameter	Design Value / Specification
Basic Wind Speed, Vb	[from Project Location]
Terrain Type	Plain Terrain
Average Exposed Height, H (m)	10 m
Hourly Mean Wind Speed, Vz	
Hourly Wind Pressure, Pz	
Transverse Wind Force	
Longitudinal Wind Force	
Vertical Wind Force	

Table 3.6: Earthquake Load (EL) — per IRC 6

Parameter	Design Value / Specification
Seismic Zone	[from Project Location]
Zone Factor, Z	
Importance Factor, I	1.0
Type of Soil	Type I – Rocky or Hard
Sa/g	
Horizontal Seismic Coefficient, Ah	
Vertical Seismic Coefficient, Av	
Horizontal Seismic Force (longitudinal)	Not available
Horizontal Seismic Force (transverse)	Not available

Table 3.7: Temperature Load (TL) — per IRC 6

Parameter	Design Value / Specification
Maximum Shade Temperature	°C
Minimum Shade Temperature	°C
Effective Bridge Temp. Range	to °C
Temperature Rise / Fall for Design	+ °C / − °C

Table 3.8: Load Combinations

Combination ID	Load Cases
ULS-01	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.50) + WL(0.90) + TL(0.90)
ULS-02	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.15) + WL(1.50) + TL(0.90)
ULS-03	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.15) + WL(0.90) + TL(1.50)
ULS-04	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + VC(1.00)
ULS-05	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + BI(1.00)
ULS-06	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + FB(1.00)
ULS-07	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(0.20) + TL(0.50) + EQ(1.50)
ULS-08	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(0.20) + TL(0.50) + EQ(0.75)
SLS-01	DL(1.00) + SIDL(1.20/1.00) + LL(1.00) + WL(0.60) + TL(0.60)
SLS-02	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(1.00) + TL(0.60)
SLS-03	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(0.60) + TL(1.00)
SLS-04	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(0.50) + TL(0.50)
SLS-05	DL(1.00) + SIDL(1.20/1.00) + LL(0.20) + WL(0.60) + TL(0.50)
SLS-06	DL(1.00) + SIDL(1.20/1.00) + LL(0.20) + WL(0.50) + TL(0.60)
SLS-07	DL(1.00) + SIDL(1.20/1.00) + LL(0.00) + WL(0.00) + TL(0.50)

Note: All IRC 6 load combinations are auto-generated by OsdagBridge. User-defined custom combinations, if any, are appended.

4 Analysis Results

A grillage model was used for structural analysis. The deck is idealized as a grid of elastic beam elements — longitudinal members represent the composite steel girders with effective slab, and transverse members represent the slab or cross frames. This section summarizes the critical output from that analysis.

Table 4.1: Summary of Maximum Demands

Load case	Bending Moment			Shear Force			Support reactions	
	Max (kNm)	Gdr.	Loc. (m)	Max (kN)	Gdr.	Loc. (m)	Left	Right
—	—	—	—	—	—	—	—	—

Table 4.2: Reactions at Supports

Load Case	Left Support (kN)	Right Support (kN)

Table 4.3: Deflection Summary (Live Load & Total Load)

parameter	value
Deflection due to Live Load, δ_{LL}	—
Allowable Live Load Deflection (Δ_{allow})	L/800 = 37.5 mm
Live Load Deflection Check Status	—
Deflection due to Total Load, δ_{total}	—
Allowable Total Deflection (Δ_{allow})	L/600 = 50.0 mm
Total Load Deflection Check Status	—

[PLACEHOLDER: Bending Moment Envelope (Envelope ULS): Max/min BM along span. X-axis: distance from left support (m). Y-axis: Bending Moment (kN-m).]

[PLACEHOLDER: Shear Force Envelope (Envelope ULS): Max/min SF along span. X-axis: distance from left support (m). Y-axis: Shear Force (kN).]

[PLACEHOLDER: Vertical Deflection D_y (1.0 LL): Maximum deflection along span. Load Case: 1.0 LL, Combination: D_y. Nodes shown. Isometric view.]

5 Design Checks

This section presents all structural design checks performed by OsdagBridge. For each member, the demand from the governing load combination, the code-based capacity, and the utilization ratio are tabulated. All checks reference IS 800:2007 and IRC 22:2014 unless stated otherwise.

5.1 Plate Girder Design

Table 5.1: Girder Section Properties (Final Optimized / User-selected)

Girder	Property	Value
G1	Depth, D (mm)	
G1	Top Flange Width, b_f (mm)	
G1	Bottom Flange Width, b_f (mm)	
G1	Top Flange Thickness, t_f (mm)	
G1	Bottom Flange Thickness, t_f (mm)	
G1	Web Thickness, t_w (mm)	
G1	Gross Area, A (cm^2)	
G1	Moment of Inertia, I_z (cm^4)	
G1	Elastic Section Modulus, Z_{ez} (cm^3)	
G1	Plastic Section Modulus, Z_{pz} (cm^3)	
G1	Effective Slab Width, b_{eff} (mm)	
G1	Transformed Composite I_z (cm^4)	
G1	Depth to Plastic Neutral Axis (mm)	
G2	Depth, D (mm)	
G2	Top Flange Width, b_f (mm)	

Continued on next page...

Table 5.1 – Continued from previous page

Girder	Property	Value
G2	Bottom Flange Width, b_f (mm)	
G2	Top Flange Thickness, t_f (mm)	
G2	Bottom Flange Thickness, t_f (mm)	
G2	Web Thickness, t_w (mm)	
G2	Gross Area, A (cm^2)	
G2	Moment of Inertia, I_z (cm^4)	
G2	Elastic Section Modulus, Z_{ez} (cm^3)	
G2	Plastic Section Modulus, Z_{pz} (cm^3)	
G2	Effective Slab Width, b_{eff} (mm)	
G2	Transformed Composite I_z (cm^4)	
G2	Depth to Plastic Neutral Axis (mm)	
G3	Depth, D (mm)	
G3	Top Flange Width, b_f (mm)	
G3	Bottom Flange Width, b_f (mm)	
G3	Top Flange Thickness, t_f (mm)	
G3	Bottom Flange Thickness, t_f (mm)	
G3	Web Thickness, t_w (mm)	
G3	Gross Area, A (cm^2)	
G3	Moment of Inertia, I_z (cm^4)	
G3	Elastic Section Modulus, Z_{ez} (cm^3)	
G3	Plastic Section Modulus, Z_{pz} (cm^3)	

Continued on next page...

Table 5.1 – Continued from previous page

Girder	Property	Value
G3	Effective Slab Width, b_{eff} (mm)	
G3	Transformed Composite I_z (cm^4)	
G3	Depth to Plastic Neutral Axis (mm)	
G4	Depth, D (mm)	
G4	Top Flange Width, b_f (mm)	
G4	Bottom Flange Width, b_f (mm)	
G4	Top Flange Thickness, t_f (mm)	
G4	Bottom Flange Thickness, t_f (mm)	
G4	Web Thickness, t_w (mm)	
G4	Gross Area, A (cm^2)	
G4	Moment of Inertia, I_z (cm^4)	
G4	Elastic Section Modulus, Z_{ez} (cm^3)	
G4	Plastic Section Modulus, Z_{pz} (cm^3)	
G4	Effective Slab Width, b_{eff} (mm)	
G4	Transformed Composite I_z (cm^4)	
G4	Depth to Plastic Neutral Axis (mm)	
G5	Depth, D (mm)	
G5	Top Flange Width, b_f (mm)	
G5	Bottom Flange Width, b_f (mm)	
G5	Top Flange Thickness, t_f (mm)	
G5	Bottom Flange Thickness, t_f (mm)	

Continued on next page...

Table 5.1 – Continued from previous page

Girder	Property	Value
G5	Web Thickness, t_w (mm)	
G5	Gross Area, A (cm^2)	
G5	Moment of Inertia, I_z (cm^4)	
G5	Elastic Section Modulus, Z_{ez} (cm^3)	
G5	Plastic Section Modulus, Z_{pz} (cm^3)	
G5	Effective Slab Width, b_{eff} (mm)	
G5	Transformed Composite I_z (cm^4)	
G5	Depth to Plastic Neutral Axis (mm)	

Table 5.2: Girder Section Classification

	Element	Slenderness Ratio	Class Limit	Classification
G1	Top Flange			
G1	Web			
G1	Overall Section	—	—	
G2	Top Flange			
G2	Web			
G2	Overall Section	—	—	
G3	Top Flange			
G3	Web			
G3	Overall Section	—	—	
G4	Top Flange			

Continued on next page...

Table 5.2 – Continued from previous page

	Element	Slenderness Ratio	Class Limit	Classification
G4	Web			
G4	Overall Section	—	—	
G5	Top Flange			
G5	Web			
G5	Overall Section	—	—	

Note: IS 800:2007 Table 2

Table 5.3: Moment Capacity Check

	Parameter	Formula	Value	Status
G1	Applied Moment, M_u	Governing LC (ULS)		—
G1	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1		—
G1	Utilization Ratio, M_u/M_d	M_u/M_d		—
G2	Applied Moment, M_u	Governing LC (ULS)		—
G2	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1		—
G2	Utilization Ratio, M_u/M_d	M_u/M_d		—
G3	Applied Moment, M_u	Governing LC (ULS)		—
G3	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1		—

Continued on next page...

Table 5.3 – Continued from previous page

	Parameter	Formula	Value	Status
G3	Utilization Ratio, M_u/M_d	M_u/M_d		—
G4	Applied Moment, M_u	Governing LC (ULS)		—
G4	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1		—
G4	Utilization Ratio, M_u/M_d	M_u/M_d		—
G5	Applied Moment, M_u	Governing LC (ULS)		—
G5	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1		—
G5	Utilization Ratio, M_u/M_d	M_u/M_d		—

Note: IRC 22 Cl. 603.3.1, IS 800 Cl. 8.2.1

Table 5.4: Shear Capacity Check

	Parameter	Formula	Value	Status
G1	Applied Shear, V_u	Governing LC (ULS)		—
G1	Shear Area, A_v	$d_w \times t_w$		—
G1	Panel Aspect Ratio, c/d	—		—
G1	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2		—
G1	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$		—

Continued on next page...

Table 5.4 – Continued from previous page

	Parameter	Formula	Value	Status
G1	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2		—
G1	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$		—
G1	Utilization Ratio, V_u/V_d	V_u/V_d		—
G2	Applied Shear, V_u	Governing LC (ULS)		—
G2	Shear Area, A_v	$d_w \times t_w$		—
G2	Panel Aspect Ratio, c/d	—		—
G2	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2		—
G2	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$		—
G2	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2		—
G2	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$		—
G2	Utilization Ratio, V_u/V_d	V_u/V_d		—
G3	Applied Shear, V_u	Governing LC (ULS)		—
G3	Shear Area, A_v	$d_w \times t_w$		—
G3	Panel Aspect Ratio, c/d	—		—
G3	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2		—

Continued on next page...

Table 5.4 – Continued from previous page

	Parameter	Formula	Value	Status
G3	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$		—
G3	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2		—
G3	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$		—
G3	Utilization Ratio, V_u/V_d	V_u/V_d		—
G4	Applied Shear, V_u	Governing LC (ULS)		—
G4	Shear Area, A_v	$d_w \times t_w$		—
G4	Panel Aspect Ratio, c/d	—		—
G4	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2		—
G4	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$		—
G4	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2		—
G4	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$		—
G4	Utilization Ratio, V_u/V_d	V_u/V_d		—
G5	Applied Shear, V_u	Governing LC (ULS)		—
G5	Shear Area, A_v	$d_w \times t_w$		—
G5	Panel Aspect Ratio, c/d	—		—

Continued on next page...

Table 5.4 – Continued from previous page

	Parameter	Formula	Value	Status
G5	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2		—
G5	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$		—
G5	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2		—
G5	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$		—
G5	Utilization Ratio, V_u/V_d	V_u/V_d		—

Note: IS 800 Cl. 8.4, IRC 22 Cl. 603.3.3.2

Table 5.5: Interaction Checks (M-V and M-N)

	Check	Condition	Value	Status
G1	High Shear Condition?	$V_u > 0.6\,V_d$		—
G1	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3		—
G1	Interaction Check: $M_u \leq M_{dv}$	—		—
G1	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	N/A	N/A	N/A
G2	High Shear Condition?	$V_u > 0.6\,V_d$		—
G2	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3		—
G2	Interaction Check: $M_u \leq M_{dv}$	—		—

Continued on next page...

Table 5.5 – Continued from previous page

	Check	Condition	Value	Status
G2	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	N/A	N/A	N/A
G3	High Shear Condition?	$V_u > 0.6\,V_d$		—
G3	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3		—
G3	Interaction Check: $M_u \leq M_{dv}$	—		—
G3	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	N/A	N/A	N/A
G4	High Shear Condition?	$V_u > 0.6\,V_d$		—
G4	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3		—
G4	Interaction Check: $M_u \leq M_{dv}$	—		—
G4	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	N/A	N/A	N/A
G5	High Shear Condition?	$V_u > 0.6\,V_d$		—
G5	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3		—
G5	Interaction Check: $M_u \leq M_{dv}$	—		—
G5	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	N/A	N/A	N/A

Note: IRC 22 Cl. 603.3.3.3

Table 5.6: Lateral Torsional Buckling Check – Construction Stage

	Parameter	Formula	Value	Status
G1	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1		—
G1	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$		—
G1	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2		—
G1	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$		—
G1	$M_u \leq M_b$	M_u/M_b		—
G2	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1		—
G2	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$		—
G2	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2		—
G2	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$		—
G2	$M_u \leq M_b$	M_u/M_b		—
G3	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1		—
G3	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$		—
G3	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2		—
G3	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$		—
G3	$M_u \leq M_b$	M_u/M_b		—
G4	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1		—

Continued on next page...

Table 5.6 – Continued from previous page

	Parameter	Formula	Value	Status
G4	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$		—
G4	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2		—
G4	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$		—
G4	$M_u \leq M_b$	M_u/M_b		—
G5	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1		—
G5	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$		—
G5	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2		—
G5	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$		—
G5	$M_u \leq M_b$	M_u/M_b		—

Note: IRC 22 Cl. 603.3.3.1, IS 800 Cl. 8.2.2

Table 5.7: Stiffener Design Summary

Girder	Property	Value
G1	Shear Buckling Design Method	
G1	Intermediate Stiffener Thickness (mm)	
G1	Intermediate Stiffener Spacing (mm)	
G1	End Panel Stiffener Thickness (mm)	
G1	No. of End Panel Stiffeners	

Continued on next page...

Table 5.7 – Continued from previous page

Girder	Property	Value
G1	Longitudinal Stiffeners	
G2	Shear Buckling Design Method	
G2	Intermediate Stiffener Thickness (mm)	
G2	Intermediate Stiffener Spacing (mm)	
G2	End Panel Stiffener Thickness (mm)	
G2	No. of End Panel Stiffeners	
G2	Longitudinal Stiffeners	
G3	Shear Buckling Design Method	
G3	Intermediate Stiffener Thickness (mm)	
G3	Intermediate Stiffener Spacing (mm)	
G3	End Panel Stiffener Thickness (mm)	
G3	No. of End Panel Stiffeners	
G3	Longitudinal Stiffeners	
G4	Shear Buckling Design Method	
G4	Intermediate Stiffener Thickness (mm)	
G4	Intermediate Stiffener Spacing (mm)	
G4	End Panel Stiffener Thickness (mm)	
G4	No. of End Panel Stiffeners	
G4	Longitudinal Stiffeners	
G5	Shear Buckling Design Method	

Continued on next page...

Table 5.7 – Continued from previous page

Girder	Property	Value
G5	Intermediate Stiffener Thickness (mm)	
G5	Intermediate Stiffener Spacing (mm)	
G5	End Panel Stiffener Thickness (mm)	
G5	No. of End Panel Stiffeners	
G5	Longitudinal Stiffeners	

Table 5.8: End Panel Stiffener Checks

	Check	Required	Provided	Status
G1	Web Buckling Resistance	N/A	N/A	N/A
G1	Local Crushing Resistance	N/A	N/A	N/A
G1	Bearing Capacity	N/A	N/A	N/A
G1	Column Buckling Resistance	N/A	N/A	N/A
G2	Web Buckling Resistance	N/A	N/A	N/A
G2	Local Crushing Resistance	N/A	N/A	N/A
G2	Bearing Capacity	N/A	N/A	N/A
G2	Column Buckling Resistance	N/A	N/A	N/A
G3	Web Buckling Resistance	N/A	N/A	N/A

Continued on next page...

Table 5.8 – Continued from previous page

	Check	Required	Provided	Status
G3	Local Crushing Resistance	N/A	N/A	N/A
G3	Bearing Capacity	N/A	N/A	N/A
G3	Column Buckling Resistance	N/A	N/A	N/A
G4	Web Buckling Resistance	N/A	N/A	N/A
G4	Local Crushing Resistance	N/A	N/A	N/A
G4	Bearing Capacity	N/A	N/A	N/A
G4	Column Buckling Resistance	N/A	N/A	N/A
G5	Web Buckling Resistance	N/A	N/A	N/A
G5	Local Crushing Resistance	N/A	N/A	N/A
G5	Bearing Capacity	N/A	N/A	N/A
G5	Column Buckling Resistance	N/A	N/A	N/A

Note: IS 800 Cl. 8.4.2.2

Table 5.9: Serviceability – Deflection Checks

	Check	Allowable	Actual	Status
G1	Live Load Deflection, δ_{LL} (mm)	L/800 = 37.5 mm		—
G1	Total Load Deflection, δ_{total} (mm)	L/600 = 50.0 mm		—
G2	Live Load Deflection, δ_{LL} (mm)	L/800 = 37.5 mm		—
G2	Total Load Deflection, δ_{total} (mm)	L/600 = 50.0 mm		—
G3	Live Load Deflection, δ_{LL} (mm)	L/800 = 37.5 mm		—
G3	Total Load Deflection, δ_{total} (mm)	L/600 = 50.0 mm		—
G4	Live Load Deflection, δ_{LL} (mm)	L/800 = 37.5 mm		—
G4	Total Load Deflection, δ_{total} (mm)	L/600 = 50.0 mm		—
G5	Live Load Deflection, δ_{LL} (mm)	L/800 = 37.5 mm		—
G5	Total Load Deflection, δ_{total} (mm)	L/600 = 50.0 mm		—

Note: IRC 22 Cl. 604.3.2

Table 5.10: Serviceability – Maximum Stress Limitation

	Element	Allowable Stress	Actual Stress	Status
G1	Structural Steel ($0.9\,f_y$)			—
G2	Structural Steel ($0.9\,f_y$)			—
G3	Structural Steel ($0.9\,f_y$)			—
G4	Structural Steel ($0.9\,f_y$)			—
G5	Structural Steel ($0.9\,f_y$)			—

Table 5.11: Serviceability – Fatigue Assessment

	Stress Range, $\Delta\sigma$ (MPa)	Fatigue Limit, f_{fd} (MPa)	Utilization Ratio	Status
G1				—
G2				—
G3				—
G4				—
G5				—

Note: IRC 22 Cl. 605 — governing of normal and shear fatigue (worst by DCR). Capacity reduction factor μ_r applied where plate thickness ¿ 25 mm.

Table 5.12: Girder Design Summary (DCR / Utilization Ratio)

Girder	Controlling LC / Combination	Controlling Check	Demand	Capacity	UR	Status
G1		Moment	121.00 kN-m	100.00 kN-m	1.210	—
G2		Moment	122.00 kN-m	100.00 kN-m	1.220	—
G3		Moment	123.00 kN-m	100.00 kN-m	1.230	—
G4		Moment	124.00 kN-m	100.00 kN-m	1.240	—
G5		Moment	125.00 kN-m	100.00 kN-m	1.250	—

Note: UR = Demand / Capacity. A value ≤ 1.0 *indicates a passing check. The controlling check is the criterion with the highest UR for each girder, with the real load case/combination that drives it.*

Table 5.13: Shear Connector Capacity

Parameter	Formula	Value	Reference
Design Resistance, Q_u	$Q_u = \min(Q_{u,s},\, Q_{u,c})$ $Q_{u,s} = \dfrac{0.8\, f_u\, (\pi d^2/4)}{\gamma_v}$ $Q_{u,c} = \dfrac{0.29\, \alpha\, d^2 \sqrt{f_{ck}\, E_{cm}}}{\gamma_v}$		IRC 22 Cl. 606.3.1 (Eq. 6.1)
Fatigue Shear Resistance, Q_r	IRC 22 Table 8 (ϕd, N_{sc})		IRC 22 Cl. 606.3.2 (Table 8)

Table 5.14: Shear Connector Spacing

Criterion	Governing Spacing	Actual Spacing Provided	Status
ULS Shear (SL1)			—
Full Composite (SL2)			—
SLS Fatigue (SR)			—
Max Spacing Limit (IRC 22)			—

Note: IRC 22 Cl. 606.4, 606.9. Governing spacing $= \min(S_{L1}, S_{L2}, S_R)$.

Table 5.15: Transverse Shear and Detailing Checks

Check	Value	Status
Longitudinal Shear per unit length, V_L		—
Transverse Shear Capacity of Slab, V_{Rd}		—
Transverse Shear Check	$V_L/V_{Rd} =$ —	—
Min. Transverse Reinforcement, $A_{st,min}$	Required , Provided 0.00 cm^2/m	—
Stud Diameter $\leq 2\,t_f$	$d = 20.0$ mm $\leq 2t_f =$	—
Stud Edge Distance	Provided (req. $\geq$)	—

Note: IRC 22 Cl. 606.6, 606.10.

5.2 Deck Slab Design

The reinforced concrete deck slab is designed per IRC 112:2011 (flexure, shear, crack width) and IRC 22:2014 (composite construction). Wheel loads are distributed using Pigeaud's method. The deck is checked for flexure in the transverse and longitudinal directions, punching shear, one-way (beam) shear, crack width, and reinforcement detailing.

Table 5.16: Deck Slab — Loading and Geometry

Effective Span of Deck Slab, l_{eff}	<—> mm (girder spacing, c/c)
Deck Thickness, t_s	250.0 mm
Clear Cover (IRC 112 Cl. 15.2)	Top 50 / Bottom 50 mm
Concrete Grade (IRC 112 Cl. 6.4)	M40 (f_{ck} = MPa, f_{ctm} = MPa)
Reinforcement Grade (IRC 112 Cl. 6.2)	Fe 500 (f_y = <—> MPa)
Dead Load per Unit Area, w_{DL}	<—> kN/m^2 (slab self-weight)
IRC 6 Wheel Load (Class A / 70R)	<—> kN
Tyre Contact Width (IRC 6 Annex A)	<—> mm (transverse)
Impact Factor (IRC 6 Cl. 208.2)	<—>
Governing Live Load Case	<—>

Table 5.17: Deck Slab — Flexure Check: Interior Panel (Pigeaud's Method)

Location	Parameter	Formula / Reference	Value	Status
At Midspan (Sagging)	Transverse BM (DL), $M_{T,DL}$	$w_{DL}\, l_{eff}^2/10$	<—> kN-m/m	—
At Midspan (Sagging)	Transverse BM (LL), $M_{T,LL}$	Effective width (IRC 112 B3.1)	<—> kN-m/m	—
At Midspan				

Continued on next page...

Table 5.17 – Continued from previous page

Location	Parameter	Formula / Reference	Value	Status
(Sagging)	Total Design BM, $M_{u,sag}$	<—> DL + <—> LL	65.00 kN-m/m	—
At Midspan (Sagging)	Effective depth, d	$t_s - c_{nom} - \phi/2$	<—> mm	—
At Midspan (Sagging)	Moment Capacity, M_{Rd}	IRC 112 Cl. 12.2	85.00 kN-m/m	PASS
At Support (Hogging)	Total Design BM, $M_{u,hog}$	<—> DL + <—> LL (at support)	<—> kN-m/m	—
At Support (Hogging)	Required Top Steel, $A_{st,top}$	$M_u/(0.87\, f_y\, d)$	<—> mm^2/m	—
At Support (Hogging)	Moment Capacity, M_{Rd}	IRC 112 Cl. 12.2	<—> kN-m/m	Not available

Note: IRC 112 Cl. 12.2. Distribution (longitudinal) reinforcement designed for 20% of main steel moment (IRC 21 Cl. 305.18).

Table 5.18: Deck Slab — Cantilever Overhang Flexure Check

Parameter	Formula	Value	Status
Overhang Length, l_{oh}	—	1.215 m	—
Crash Barrier Load Moment	IRC 6 Cl. 206.4	N/A	—
Dead Load Moment	$w_{DL}\, l_{oh}^2/2$ + railing	N/A	—
Live Load Moment (eccentric wheel)	Wheel load × arm	N/A	—
Total Hogging Moment, $M_{u,oh}$	<—> DL + <—> (LL + CB)	N/A	—
Moment Capacity (top steel), $M_{Rd,oh}$	IRC 112 Cl. 12.2	N/A	N/A

Note: IRC 6 Cl. 206.4 crash barrier loads applied at kerb face; IRC 112 Cl. 12.2 flexure.

Table 5.19: Deck Slab — Punching Shear Check (IRC 112 Cl. 10.4.6)

Parameter	Formula / Reference	Value	Status
Design Wheel Load (ULS), V_{Ed}	$\gamma_Q\,(1+IF)\,P_w$	<—> kN	—
Tyre Contact Area	$a \times b$ (IRC 6 Annex A)	<—> × <—> mm	—
Loaded Area at mid-depth, b_0	$c_1 \times c_2$ (incl. WC dispersion)	<—> × <—> mm	—
Control Perimeter, u_1	$2(c_1 + c_2) + 4\pi d$	<—> mm	—
Punching Shear Stress, v_{Ed}	$V_{Ed}/(u_1\, d)$	<—> MPa	—
Punching Resistance, $v_{Rd,c}$	IRC 112 Eq. 10.1	<—> MPa	—
Punching Shear Check	$v_{Ed} \le v_{Rd,c}$	<—>	Not available

Note: Punching shear reinforcement not typically required for deck slabs with $d \ge 200$ mm and adequate longitudinal reinforcement.

Table 5.20: Crack Width Check (Deck Slab)

Parameter	Value / Reference
Min. Reinforcement for Crack Control, $A_{s,min}$	<—> mm²/m [IRC 112 Cl. 16.5.1]
Provided Reinforcement (bottom)	ϕ<—> @ <—> mm c/c (<—> mm²/m)
Max. Permissible Crack Width	<—> mm
Calculated Crack Width, w_k (governing)	<—> mm
Crack Width Check	Not available

Table 5.21: One-Way (Beam) Shear Check (Deck Slab)

Parameter	Formula / Reference	Value	Status
Design Shear per unit width, V_{Ed}	$\gamma_{DL}V_{DL} + \gamma_{LL}(1+IF)V_{LL}$	<—> kN/m	—
Effective depth, d	$t_s - c_{nom} - \phi/2$	<—> mm	—
Size factor, k	$1 + \sqrt{200/d} \leq 2.0$	<—>	—
Long. reinforcement ratio, ρ_l	$A_{sl}/(b_w\, d) \leq 0.02$	<—>	—
Shear resistance (no stirrups), $V_{Rd,c}$	$v_{Rd,c}\, b_w\, d$ (Cl. 10.3.2)	<—> kN/m	—
One-Way Shear Check	$V_{Ed} \leq V_{Rd,c}$	<—>	Not available

Note: IRC 112 Cl. 10.3.2. Shear reinforcement not provided in deck slabs; capacity relies on concrete and main reinforcement.

Table 5.22: Reinforcement Detailing Summary (Deck Slab)

Parameter	Required / Limit	Provided	Status
Main Reinforcement — Bottom (Transverse)			
Required Area, $A_{st,req}$ (mm^2/m)	<—> mm^2/m	<—> mm^2/m	Not available
Bar Diameter × Spacing	$\phi \geq 10$ mm (IRC 112)	ϕ<—> @ <—> mm c/c	—
Min. Reinforcement $A_{s,min}$ (IRC 112 Cl. 16.3.1)	<—> mm^2/m	<—> mm^2/m	Not available
Max. Bar Spacing (IRC 112 Cl. 16.3.2)	<—> mm	<—> mm	Not available
Distribution Reinforcement — Longitudinal			
Required Area, $A_{st,dist}$ (mm^2/m)	$\geq$ 20% of main steel	<—> mm^2/m	Not available
Top Reinforcement (Support / Cantilever Overhang)			
Required Area, $A_{st,top}$ (mm^2/m)	<—> mm^2/m	<—> mm^2/m	Not available
Cover and Detailing			
Clear Cover (IRC 112 Cl. 15.2)	$\geq$ <—> mm (Table 14.2)	Top 50 / Bottom 50 mm	Not available

Note: IRC 112 Cl. 16.3, IS 456 Cl. 26.5. All reinforcement provisions satisfy strength and detailing requirements.

5.3 Cross Bracing Design

Cross bracing between adjacent plate girders provides lateral stability during construction, resists transverse loads (wind, seismic, braking) in service, and prevents lateral torsional buckling of the girders. Members are designed per IS 800:2007 Cl. 7 (compression) and Cl. 6 (tension). Forces are derived from the grillage model under the governing load combination (DL + LL + WL).

Table 5.23: Cross Bracing — Connection and Section Properties

Panel	Member	Connection	Section	A_g (mm²)	r_{min} (mm)
Between Girders	Diagonal				

Note: A_g = gross cross-sectional area; r_{min} = minimum radius of gyration.

Table 5.24: Cross Bracing — Slenderness Ratio Check (IS 800 Cl. 3.8)

Panel	Member	Nature	Eff. Length KL (mm)	KL/r	Limit / Status
Between Girders	Diagonal	C			—

Note: 3. Limit = 250 for compression members, 400 for tension members. K = 1.0 for members with both ends pinned.

Table 5.25: Cross Bracing Design — Capacity Summary

Panel	Member	Section	Gov. LC	Demand (kN)	Capacity (kN)	UR	Status

Note: Designed per IS 800 Cl. 7 (compression) and Cl. 6 (tension). OsdagBridge cross-bracing module used.

5.4 End Diaphragm Design

End diaphragms at the supports transfer transverse loads to the bearings, restrain the bottom flanges against lateral displacement, and maintain the girder cross-section geometry during construction and in service. They are designed per IS 800:2007 and IRC 24:2010 Cl. 507.

Table 5.26: End Diaphragm — Connection and Section Properties

Panel	Member	Connection	Section	A_g (mm²)	r_{min} (mm)
Not available: verified end-diaphragm results are not supplied.					

Note: A_g = gross cross-sectional area; r_{min} = minimum radius of gyration.

Table 5.27: End Diaphragm — Slenderness Ratio Check (IS 800 Cl. 3.8)

Panel	Member	Nature	Eff. Length KL (mm)	KL/r	Limit / Status
Not available: verified end-diaphragm results are not supplied.					

Note: 3. Limit = 250 for compression members, 400 for tension members. K = 1.0 for members with both ends pinned.

Table 5.28: End Diaphragm Design — Capacity Summary

Panel	Member	Section	Gov. LC	Demand (kN)	Capacity (kN)	UR	Status
Not available: verified end-diaphragm results are not supplied.							

Note: No design status is inferred from generic cross-bracing results.

5.5 Overall Design Check Summary

This section synthesizes the critical limit state utilization ratios (UR = Demand/Capacity) governing primary superstructure elements: Steel Plate Girders, Concrete Deck Slab, Cross Bracing, and End Diaphragms. Utilization ratios at or below 1.0 indicate compliance with IRC and IS limit state requirements.

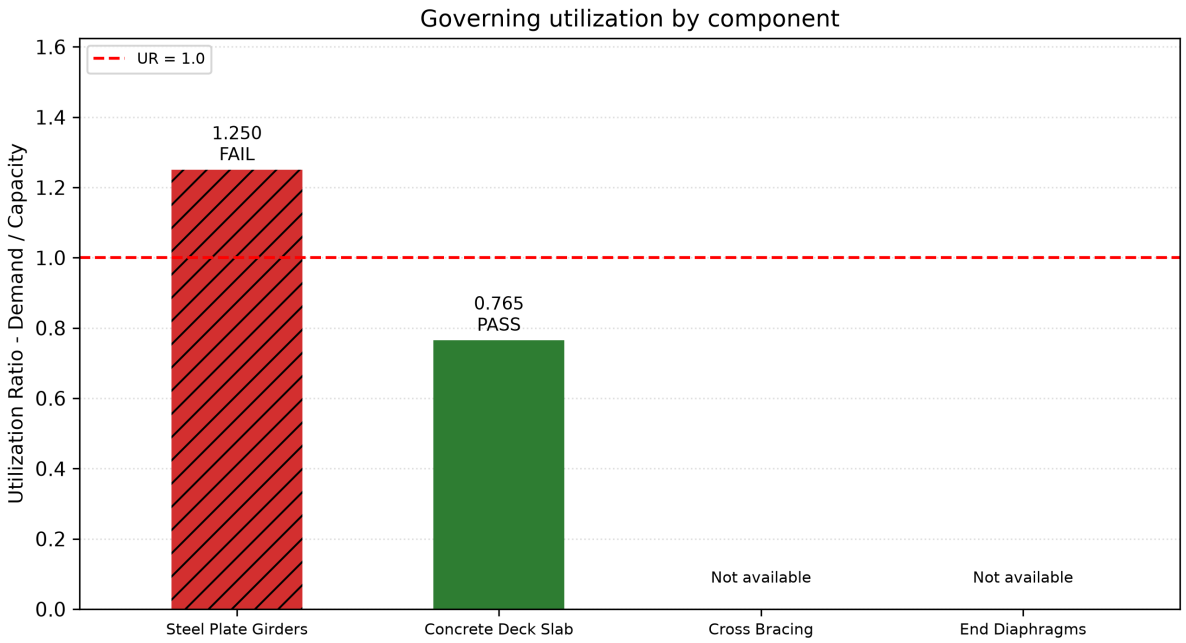


Figure 5.1: Utilization Ratio (Demand / Capacity) Summary for Primary Bridge Elements (Steel Plate Girders, Concrete Deck Slab, Cross Bracing, End Diaphragms)

Table 5.29: Overall Design Check Summary — All Members

Member / Check	Governing Load Combo	Demand	Capacity	UR
Steel plate girders / G5 / Moment	Envelope	125.000 kN-m	100.000 kN-m	1.250 (FAIL)
Concrete deck slab / Deck / Flexure sagging	Basic ULS	65.000 kN-m/m	85.000 kN-m/m	0.765 (PASS)
Cross bracing / - / Backend result unavailable	Not available	Not available	Not available	Not available
End diaphragms / - / Backend result unavailable	Not available	Not available	Not available	Not available

Note: UR = Demand / Capacity. All values ≤ 1.0 indicate passing checks. The governing check for each component is highlighted in the individual design check sections above.

6 Drawings and Visualizations

This section presents CAD-generated views of the designed bridge and its components. All views are generated automatically by OsdagBridge using pythonOCC.

6.1 Bridge Configuration and Layout

[PLACEHOLDER: Overall 3D Bridge Superstructure]

Figure 6.1: Overall 3D Bridge Superstructure

[PLACEHOLDER: Typical Cross Section]

Figure 6.2: Typical Cross Section

[PLACEHOLDER: Top View]

Figure 6.3: Top View

6.2 Plate Girder — Detailed Views

[PLACEHOLDER: 3D View of Plate Girders]

Figure 6.4: 3D View of Plate Girders

[PLACEHOLDER: Cross Section of Plate Girder]

Figure 6.5: Cross Section of Plate Girder

[PLACEHOLDER: Side View of Girder]

Figure 6.6: Side View of Girder

6.3 Cross Bracing Detail

[PLACEHOLDER: Cross Bracing Layout]

Figure 6.7: Cross Bracing Layout

[Cross Bracing – Bracing Section]

Figure 6.8: Cross Bracing – Bracing Section

[Cross Bracing – Top Chord Section]

Figure 6.9: Cross Bracing – Top Chord Section

[Cross Bracing – Bottom Chord Section]

Figure 6.10: Cross Bracing – Bottom Chord Section

6.4 End Diaphragm Detail

[PLACEHOLDER: End Diaphragm Layout]

Figure 6.11: End Diaphragm Layout

[End Diaphragm – Bracing Section]

Figure 6.12: End Diaphragm – Bracing Section

[End Diaphragm – Top Chord Section]

Figure 6.13: End Diaphragm – Top Chord Section

[End Diaphragm – Bottom Chord Section]

Figure 6.14: End Diaphragm – Bottom Chord Section

7 Material Take-off & Quantity Summary

This chapter details the comprehensive Bill of Materials (BOM) for all structural components, including structural steel (IS 2062), deck concrete (M40), reinforcement steel (Fe 500), and shear connector studs.

Table 7.1: Bill of Materials (Steel, Concrete, and Reinforcement Quantities)

No.	Item Description	Vol.	Qty.	Total Volume	Wt. (MT)	Total Wt. (MT)
1	Structural Steel (IS 2062) for Girders	0.03492 m^2 × 30.00 m = 1.04756 m^3	5	5.24	8.22	41.12
2(a)	Cross Bracing - Top Chord	N.A.	N.A.	N.A.	N.A.	N.A.
2(b)	Cross Bracing - Bottom Chord	N.A.	N.A.	N.A.	N.A.	N.A.
2(c)	Cross Bracing - Diagonal Chord	N.A.	N.A.	N.A.	N.A.	N.A.
3	Concrete (M40) for Deck Slab	12.15 m × 0.25 m × 30.00 m = 91.12 m^3	1	91.12	227.81	227.81
4	Reinforcement Steel (Fe 500)	0.046433 m^2 × 30.00 m = 1.39299 m^3	1	1.39	10.93	10.93
5	Shear Stud Connectors	N.A.	N.A.	N.A.	N.A.	N.A.
6	Crash Barrier	0.30228 m^2 × 30.00 m = 9.06848 m^3	2	18.14	22.67	45.34

Reinforcement weight is an estimate using 120 kg per cubic metre of concrete. End-diaphragm quantity is not available from the backend.

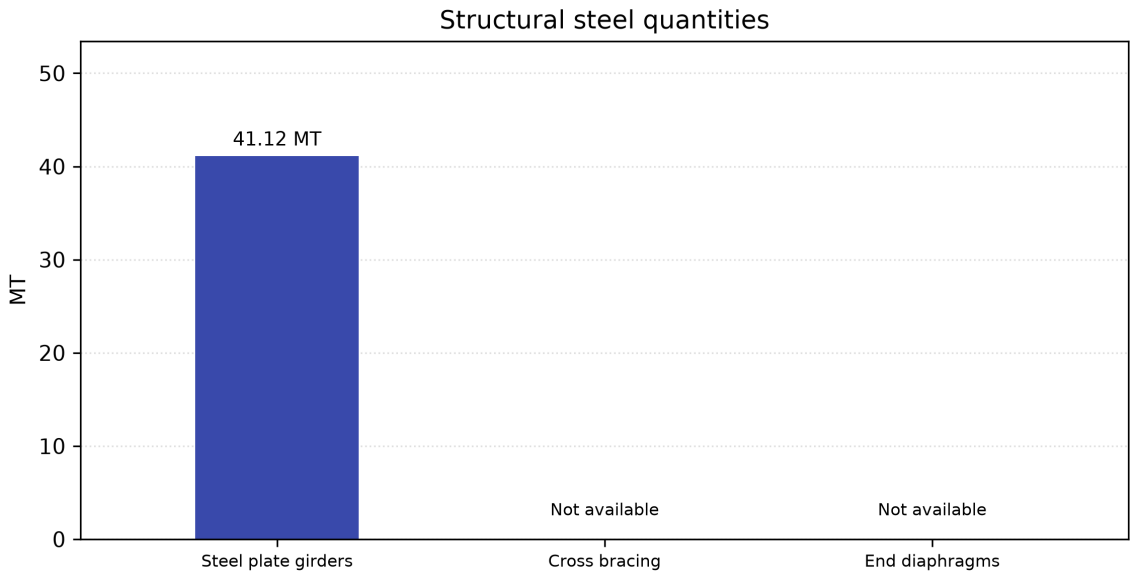


Figure 7.1: Structural Steel Tonnage Breakdown by Component (MT) — Girders, Cross Bracing, End Diaphragms

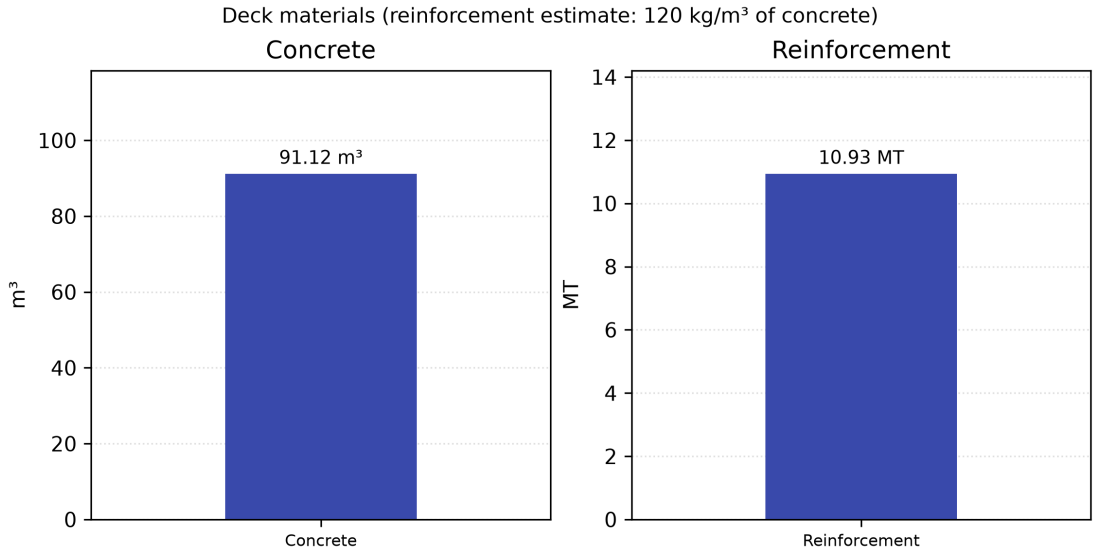


Figure 7.2: Primary Deck Materials: Concrete Volume (m^3) vs. Reinforcement Steel Weight (MT) Take-off

8 Standards & Assumptions

This section provides references to standards used to calibrate the OsdagBridge design modules, and notes the limitations of the current software version.

8.1 Design Standards

The following Indian Road Congress (IRC) codes and Indian Standards (IS) form the basis of all design calculations in this software.

Table 8.1: IRC Codes

Code	Year	Title / Scope
IRC 5	2015	General Features of Design - carriageway widths, kerb, footpath dimensions
IRC 6	2017	Loads and Load Combinations - dead load, live load, impact, wind, temperature, etc.
IRC 22	2015	Composite Construction (LS) - Composite section properties, ULS/SLS design, shear connectors
IRC 24	2010	Steel Road Bridges (LS) - Stiffener design, skew angle limits, diaphragm requirements
IRC 112	2020	Concrete Road Bridges - Deck slab flexure, shear, crack width, reinforcement
IRC SP 114	2018	Seismic Design of Road Bridges

Table 8.2: IS Codes

Code	Year	Scope
IS 800	2007	Steel construction - tension, compression, bending, shear, LTB, stiffeners, combined checks
IS 456	2000	Concrete - simplified stress-block for deck moment capacity
IS 1786	2008	Reinforcement steel properties
IS 1893 (Part 3)	2014	Earthquake resistant design
IS 2062	2011	Structural steel - yield and ultimate strength by grade

8.2 Analysis and Design Assumptions of This Version

Structural Analysis

- All girders are modelled as simply supported; continuous spans are not currently supported.
- A 3D grillage model (OSPGrillage) is used for load distribution. Grillage members carry composite section properties after the construction stage.
- The transverse member forces are computed using an approximate 2D frame analogy. For irregular or skewed geometries, a 3D FEM is recommended.
- Fixed bearing stiffness is modelled as $k = 1{,}000{,}000\,\text{kN/m}$ (virtually rigid); free/expansion bearing as $k = 100\,\text{kN/m}$.
- Construction stage sequence analysis is approximate. Detailed staged analysis should be performed for long-term deflection checks.

Composite Action

- Full shear connection is assumed at ULS with headed stud connectors designed per IRC 22:2015 Cl.606.
- Short-term composite section properties (modular ratio $n = E_s/E_{cm}$) are used for ULS checks.
- Long-term composite section properties (with creep-adjusted modular ratio) are used for SLS deflection and crack-width checks.
- Pre-composite stage: the steel girder alone resists all construction loads prior to concrete gaining strength.

Material Properties (IRC 22:2015 Annex III)

- Steel: $E_s = 200{,}000\,\text{MPa}$, $G_s = 80{,}000\,\text{MPa}$, $\nu = 0.30$, $\alpha = 11.7 \times 10^{-6}/°\text{C}$ (grade-independent).
- Minimum structural concrete grade: M25.
- Default reinforcement grade: Fe500.

Partial Safety Factors (IRC 22:2015 Cl.601.4)

- γ_{m0} (steel yield, ULS) = 1.10
- γ_{m1} (steel ultimate, ULS) = 1.25
- Reinforcement (ULS) = 1.15
- Welds – shop: 1.25; field: 1.50
- Fatigue (γ_{mft}) = 1.35

Loading

- Dead load densities per IRC 6:2017 Cl.203: structural steel 78.5 kN/m^3, concrete 25.0 kN/m^3, bituminous wearing course 24.0 kN/m^3.
- Multi-lane live load reduction factors per IRC 6:2017 Cl.204.4 Table 6A: 1st lane = 1.0, 2nd lane = 0.8, 3rd lane onwards = 0.4.
- Impact factor (dynamic load allowance) computed from span per IRC 6:2017 Cl.208.2/ 208.3.
- Wind load applied as transverse, longitudinal, and vertical components per IRC 6:2017 Cl.209.3.3–209.3.5.

Serviceability Limits

- Deflection limits (IRC 22:2015 Cl.604.3.2): live load + impact $\leq L/800$; total $\leq L/600$.
- SLS stress limits: concrete $\sigma_c \leq 0.48 f_{ck}$; rebar $\sigma_s \leq 0.80 f_{yk}$; steel $f_e \leq 0.9 f_y$.
- Permissible crack width: $w_k \leq 0.3$ mm (bridge deck, exposure class XS2/XD2 per IRC 112:2020 Cl.12.3.2).

Fatigue

- Reference fatigue life: $N_{sc} = 2 \times 10^6$ cycles (IRC 22:2015 Cl.605).
- Constant stress range is assumed. A thickness correction factor μ_r is applied for plate thickness > 25 mm.

Stiffener Design

- Intermediate transverse stiffener and bearing stiffener design follows IS 800:2007 Cl.8.7.2/8.7.3 and IRC 24:2010 Cl.509.7.2/509.7.3.

8.3 Known Limitations of This Version

- Substructure (piers, pile caps, foundations) and bearing design are not included.
- Splice connection design is not implemented.
- Skew angle > 15 degrees requires independent manual analysis (IRC 24 Cl. 504.8).
- Construction stage sequence analysis is approximate; detailed staged analysis should be performed for long-term deflection checks.
- The grillage analysis assumes simply supported boundary conditions; continuous spans are not currently supported.

References

1. IRC 5 (2024) — *Standard Specifications and Code of Practice for Road Bridges, Section I: General Features of Design.*

2. IRC 6 (2017) — *Standard Specifications and Code of Practice for Road Bridges, Section II: Loads and Load Combinations.*

3. IRC 22 (2014) — *Standard Specifications and Code of Practice for Road Bridges, Section VI: Composite Construction (Limit State Design).*

4. IRC 24 (2010) — *Standard Specifications and Code of Practice for Road Bridges, Section V: Steel Road Bridges (Limit State Method).*

5. IRC 112 (2011) — *Code of Practice for Concrete Road Bridges.*

6. IRC SP 114 (2018) — *Guidelines for Seismic Design of Road Bridges.*

7. IS 800 (2007) — *Indian Standard: General Construction in Steel — Code of Practice.*

8. IS 2062 (2011) — *Hot Rolled Medium and High Tensile Structural Steel — Specification.*

9. Subramanian, N. (2008). *Design of Steel Structures.* Oxford University Press.

10. Steel-INSDAG Teaching Resource Materials. `https://www.steel-insdag.org`